

ESG Climate Risk in ASEAN Financial Institution

A Quantitative Assessment of Physical and Transition Risk

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Executive Summary

Malaysian banks face material climate risk that is not currently disclosed or priced. This analysis quantifies :

<i>Risk Type</i>	<i>Key Finding</i>
Physical Risk (Flood)	RM168 billion in flood-exposed loans across 3 banks (11-12% of portfolios)
Transition Risk (Carbon Price)	RM8.5 billion in additional credit losses under RM150/tonne carbon price
NGFS Stress Test	2.8% to 6.7% portfolio losses on RM100B assets across 3 climate scenarios

Three Recommendations:

- 1.Mandate geographic disclosure of loan exposures*
- 2.Implement phased carbon pricing (RM50→RM150/t by 2030)*
- 3.Integrate NETR national policies into NGFS scenarios*

Why Climate Risk Matters for Malaysian Banks

*Key Question : How much climate risk
do Malaysian banks actually hold?*

BNM CRST 2024

Banks must now conduct climate stress tests

No current disclosure

None of the 3 banks disclose flood-zone loan exposure

Global pressure

Trading partners may impose carbon border adjustments

RM billions at stake

Physical and transition risks could materially affect loan books

Regulatory Landscape

Malaysia vs Global Peers

Dimension	Malaysia	Singapore	UK/EU	Gap
Carbon pricing	✗ No (shadow only)	✗ No (tax pending)	✓ Yes	Major
TCFD mandatory	✗ Coming	✗ No	✓ Yes	Medium
Climate stress testing	✓ First cycle	✓	✓ Mature	Maturity
Taxonomy detail	Principles-based	Transition-focused	Highly detailed	Medium

Key takeaway: Malaysia has the regulatory foundation but lacks enforcement and granularity.

How Malaysian Banks Compare on Climate Readiness

Dimension	Maybank	CIMB	Public Bank
Net zero commitment	✔ Yes (2050)	✔ Yes (NZBA member)	✗ Not stated
Sustainable finance (2024)	RM46.7B	RM117B (cumulative)	Not disclosed
TCFD alignment	Partial	Partial (full by 2027)	Not evident
Impaired loans ratio	0.52%	2.1%	0.52%

***Key takeaway:** Maybank and CIMB lead on climate commitments. Public Bank lags despite strong asset quality.*

Physical Risk: RM168 Billion in Flood-Exposed Loans

Bank	Estimated Exposure	% of Loans
Maybank	RM76.7B	11.5%
Public Bank	RM49.2B	11.6%
CIMB	RM42.5B	9.4%
Total	RM168.4B	~11%

Assumptions: 15% property loans in flood zones, based on JPS flood maps and BNM CRST 2024

Key takeaway: Household mortgages (55-70% of portfolios) are the primary driver. Banks do not currently disclose geographic loan distribution.

Transition Risk: RM8.5 Billion in Additional Credit Losses

Under RM150/tonne carbon price scenario (BNM CRST 2024 benchmark)

Bank	High-Carbon Loans	ECL Impact
Maybank	RM57.0B	RM8.5B
CIMB	RM57.0B	RM8.6B
Public Bank	RM16.0B	RM2.4B

Key sectors at risk: Manufacturing, Agriculture, Utilities, Mining, Oil & Gas

Key takeaway: Manufacturing is the largest contributor to transition risk across all three banks.

NGFS Portfolio Stress Test: 2.8% to 6.7% Losses

Representative Malaysian bank portfolio (RM100B assets)

Scenario	Portfolio Loss	Loss on RM100B	Primary Risk
Net Zero 2050 (orderly)	2.78%	RM2.78B	Transition
Divergent Net Zero	5.00%	RM5.00B	Aggressive transition
NDCs (hot house world)	6.71%	RM6.71B	Physical (floods)

Key takeaway: *Early action (Net Zero 2050) minimizes total losses. No action (NDCs) results in highest losses due to compounding physical risk.*

5 Key Findings

#1

RM168B in flood-exposed loans — banks don't disclose where

#2

RM8.5B+ in carbon price-driven credit losses — but Malaysia has no actual carbon price

#3

NGFS stress test range: 2.8% to 6.7% loss on RM100B portfolio

#4

Malaysia lags Singapore/UK/EU on carbon pricing, TCFD, and scenario granularity

#5

Public Bank lags Maybank/CIMB on climate commitments despite strongest asset quality

The bottom line: Malaysian banks face material climate risk. Regulatory gaps persist. Early action reduces losses.

3 Recommendations

MANDATE GEOGRAPHIC DISCLOURE

BNM require flood-zone loan reporting

IMPLEMENT PHASED CARBON PRICING

RM50→RM150/t by 2030

INTEGRATE NETR INTO SCENARIOS

Country-specific stress tests

Addressing these gaps would significantly strengthen Malaysia's financial system resilience and align it with global climate risk management standards.

3 Horizon Roadmap

H1 : 0-12 MONTHS

Mandate disclosure,
complete CRST 2024

H2 : 1-3 YEARS

Carbon pricing, TCFD
mandatory, NETR
integration

H3 : FULL GLOBAL ALIGNMENT, CARBON MARKET

Full global alignment,
carbon market

Conclusion: Early action minimizes losses. Malaysia can lead ASEAN in climate-resilient finance.